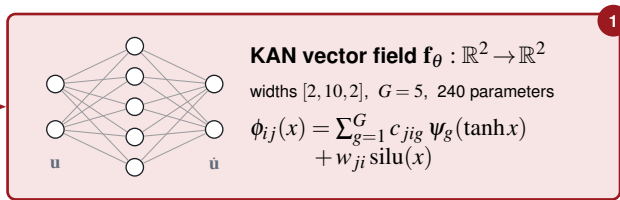
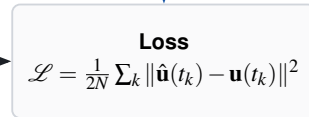
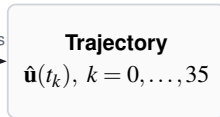
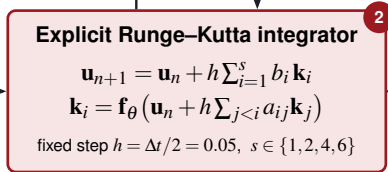


Adam: $\theta \leftarrow \theta - \eta \hat{m} / \sqrt{\hat{v}}$

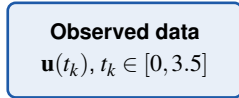


stage state
 $\mathbf{u}_n + h \sum_j a_{ij} \mathbf{k}_j$

stage slope
 $\mathbf{k}_i$



- Varied in this report**
- 1** edge basis ψ_g (7 families)
 - 2** integrator $(A, \mathbf{b}, \mathbf{c})$ (6 methods)
 - 3** gradient method (autograd vs. adjoint)



3

$\nabla_\theta \mathcal{L}$ by reverse-mode autograd through all 70 s stage evaluations (continuous adjoint: Section 4)